Fiscal Year:	FY 2023	**Task Last Updated:**	FY 04/04/2024
PI Name:	Vunjak-Novakovic, Gordana Ph.D.		
Project Title:	MORPH: Multi-Organ Repair Post Hypoxia		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	NIH, BARDA, FDA	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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PI Organization Type:	UNIVERSITY	**Phone:**	212-305-2304
Organization Name:	Columbia University		
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Zip Code:	10032	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA015N. Extended Longevity of 3D Tissues and Microphysiological Systems for Modeling of Acute and Chronic Exposures to Stressors
Start Date:	11/01/2022	**End Date:**	10/31/2026
No. of Post Docs:	4	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Per the Research Announcement for this solicitation: this effort is a collaboration between NASA; the National Institutes of Health (NIH); the Biomedical Advanced Research and Development Authority (BARDA); and the Food and Drug Administration (FDA).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Maletic-Savatic, Mirjana M.D., Ph.D. (Baylor College of Medicine, Inc.)		
Grant/Contract No.:	80ARC023CA004		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Microphysiological systems (MPS) have enabled unprecedented studies of human models of disease, holding promise to transform biomedical and medical research. Yet we need new technological advances to place this technology to the forefront of clinical utilization. To this end, we established a collaboration between Columbia University (expertise in human multi-tissue platforms) and Baylor College of Medicine (expertise in brain organoids), with collaborations with CFD Research Corporation (CFDRC, expertise in computational modeling of multi-tissue platforms, brain and heart tissues) and engaged additional expertise in the critical aspects of the project. This team will expand our existing, well-established linked multi-organ MPS (heart, bone marrow, vasculature) to include brain tissue, and establish a longitudinal pipeline for testing the functional viability of these tissues over long culture durations. Our linked multi-organ MPS platform has the capability to model the organ physiology by studying each tissue in the systemic context provided by tissue cross-talk via vascular perfusion containing circulating immune cells. New Project for FY2023. Our working hypothesis is that the multi-organ platform linked by vascular perfusion will be able to maintain matured human tissues over at least 6 months of culture, and to recapitulate and counteract acute hypoxia-induced fibrosis/gliosis. This hypothesis will be tested by pursuing three specific aims, in an integrated fashion. In Aim 1, we will demonstrate functional viability of linked tissues for up to 6 months. In Aim 2, we will examine the effects of acute hypoxia in the human multi-tissue platform. We will monitor the development of fibrosis (heart) and gliosis (brain), each linked to bone marrow by vascular perfusion containing immune cells for up to the 6-month time point, and assess the same parameters as in Aim 1. In Aim 3, we will determine if the fibrosis/gliosis following hypoxia can be counteracted by tissue-protective extracellular vesicles (EVs). Over the period of 6 months, we will monitor functional, biochemical, and morphological markers of organ health, maturation, and stress. We will continuously monitor oxygen consumption, the metabolome, cytokinome, and proteome in the media; secretion of tissue specific biomarkers; and will perform imaging of real-time calcium and oxidative status, and measure contractile behavior for heart tissue. The work will be done with the aid of longitudinal computational modeling of the transport and distributions of oxygen and EVs in tissues and vascular flow. The new tools we produce will be tested at two sites to ensure reproducibility and cross-validation. The proposed approach is designed to advance the use of linked multi-organ MPS platforms to the next level.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential Impact: Relevance to space exploration: Configurable platform for automated operation on Earth and in space. Scientific discovery: iPSC-derived human tissues of high biological fidelity linked into physiological units. Outcomes: Countermeasure for hypoxic stress, for clinical use, and space missions. Deliverables: Hardware and operating protocols for studies of bioengineered human tissues; new data and methods to drive the innovation-implementation cycle of medical research relevant to NASA missions.
Task Progress:	New Project for FY2023. 2023 UPDATE There are a number of unknown risks associated with deep space travel to the Moon and Mars, including exposure to radiation, hypoxia, and additional unknown stressors. Although human tissue models have emerged as useful predictors of human pathophysiology, a major unmet challenge in modeling human physiology in space is to enable long-term culture of physiologically relevant engineered tissues with the preservation of their phenotypes and organ-level functions. In our previous studies, we engineered functional human tissues of high biological fidelity using tissue-specific parenchymal and supporting cells derived from induced pluripotent stem cells (iPSCs). These tissues were matured to adult-like phenotypes, and evaluated longitudinally using quantitative online methods. We also established a configurable microphysiological platform for linking the bioengineered tissues to each other and allowing their crosstalk via circulatory perfusion and a selectively permeable vasculature barrier. In the present progress, we are adapting the existing platforms to extended culture of heart-bone marrow-vascular tissues (at Columbia University) and brain-bone marrow-vascular tissues (at Baylor College of Medicine). In both platform configurations, the bone marrow serves as a source of immune cells. Our goal is to enable immune cell-tissue interactions over long durations of culture (up to 6 months).
Bibliography Type:	Description: (Last Updated: 07/15/2026)
Abstracts for Journals and Proceedings	Tavakol DN, Matkovic I, Mendonca, D, Cappuccio G, Yeager K, Baldassarri I, Chesnais F, Lock RI, Teles D, Chen C, Graney P, Daly G, Maletic-Savatic M, Vunjak-Novakovic G. "Extending the lifetime of systemic, multi-organ bioengineered human tissue models with integrated immune components." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Articles in Peer-reviewed Journals	Cappuccio G, Khalil SM, Osenberg S, Li F, Maletic-Savatic M. "Mass spectrometry imaging as an emerging tool for studying metabolism in human brain organoids." Front Mol Biosci. 2023 May 22;10:1181965. https://doi.org/10.3389/fmolb.2023.1181965 ; PMID: 37304070; PMCID: PMC10251497 , May-2023